Final Presentation

Palmeiras

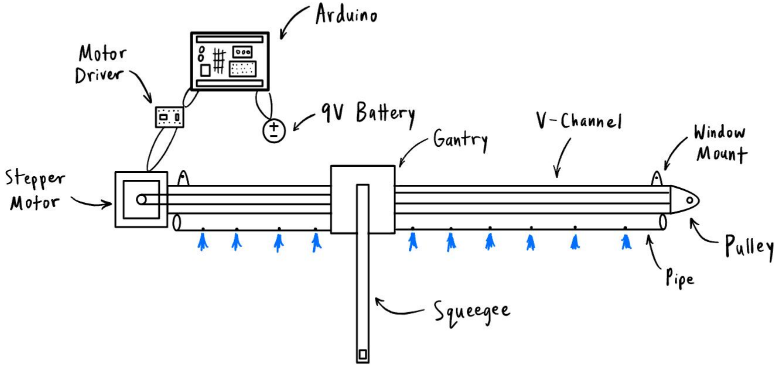
Prototype 3

Define

- Figure out the water/plumbing system (items and system)
- Make a window and mount everything on the window
- Move the gantry fully along the window

Ideate

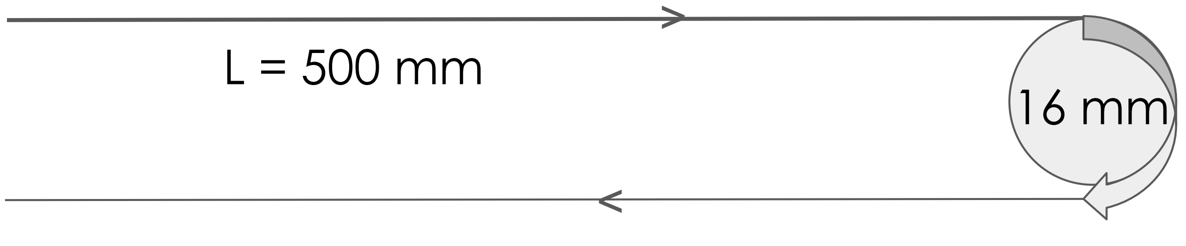
- A full length squeegee travels along length of track
- A plumbing system runs along the track, dispensing water
- All items are all attached together



Calculate

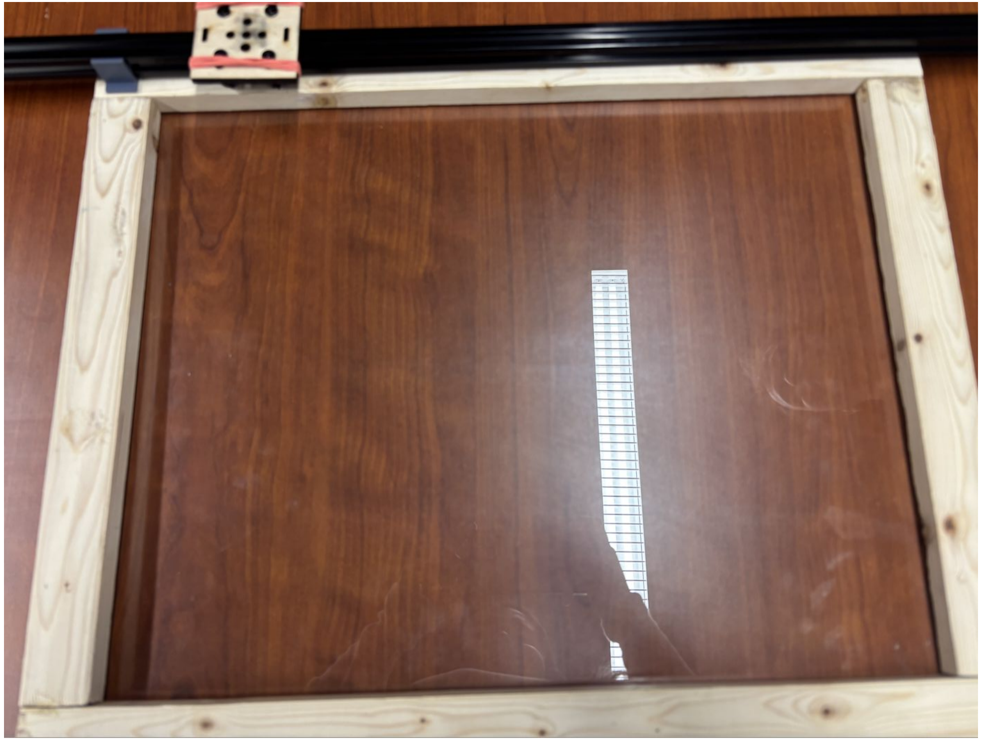
Calculated the number of rotations from the motor to cover the entire distance of the window:

- For L = 500 mm and D = 16 mm
- $N\pi D = L$
- N= Number of rotations = 9.95 rotations to cover this distance



Prototype/Test

- Ultimately, Prototype 3 mostly failed
- Had pieces, unable to put them together because of 3D printing mishaps



Learn

- Were unable to build window frame until late
  - No access to strong enough drill for screws
  - Makerspace clamps didn’t work on glue for wooden frame.
  - Frame was two small
- Not having anything to mount the prototype to, nor to test on the window the minor details of the code
- Barrel Jack adapter, limit switches not going to arrive in time
  - Didn’t have access to power supply between meetings

Fixing Prototype 3

- Immediately after the TA meeting, we were able to:
  - Re-print fasteners for window to window frame
  - Got smaller screws to build frame

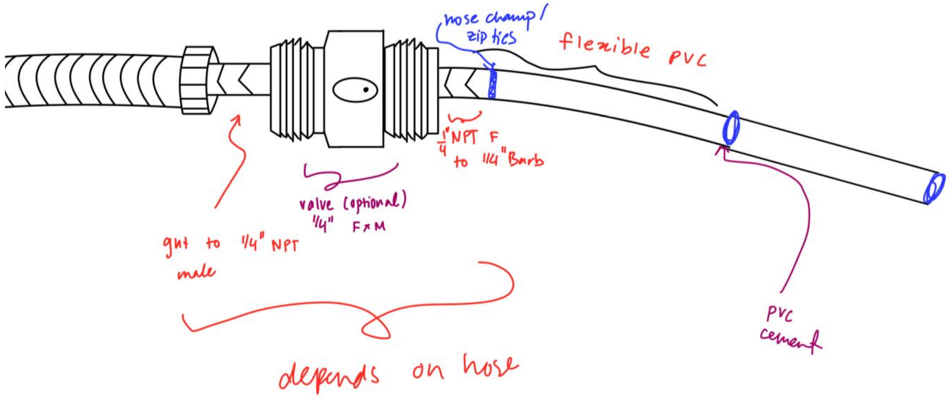
Prototype 4

Define

- Must be able to spray water across the window
- Has to have a full length squeegee spanning the height of the window
- Must be an applied force on the squeegee to improve cleaning

Ideate

- Plan to add a water dispensing system
- Will be attached to garden hose and spray water on the window before wiping motion



Calculate

Calculated the head loss throughout the pipe, to calculate power needed through the hose.

$V = \frac{Q}{A}$ $D = 0.00635 m$

$v = 1.05 \ m/s$ $Q = 2 \frac{L}{min}$

$$Re = \frac{1000 \cdot 1.05 \cdot 0.00635}{0.001} = 6680$$

$f \approx 0.035$ moody chart

$$h_f = f \frac{L}{D} \cdot \frac{v^2}{2g}$$

$h_f = 62 \ cm$

$62 cm + \underbrace{100 cm}_{\text{assumed elevation from window measurement}} + \overset{\text{minor head loss}}{12 cm}$

$= 174 \ cm$ HL

Prototype/Test

Window Built and Mounted!

Arduino Code Finalized with Limit Switches!

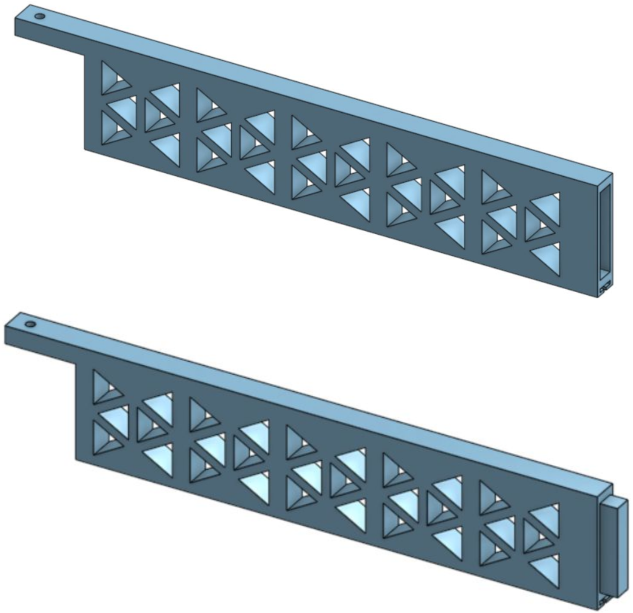
Arduino Uno

TwoSwitch.ino

```
const int stepPin = 2;
const int dirPin = 3;

const int limitSwitch1Pin = 7; // One end
const int limitSwitch2Pin = 8; // Other end

bool direction = LOW; // arb start in a direction

void setup() {
  pinMode(stepPin, OUTPUT);
  pinMode(dirPin, OUTPUT);

  pinMode(limitSwitch1Pin, INPUT_PULLUP); // N.O.
  pinMode(limitSwitch2Pin, INPUT_PULLUP); // N.O.
}

void loop() {
  digitalWrite(dirPin, direction);

  while (true) {
    // check if either limit switch is pressed
    if ((digitalRead(limitSwitch1Pin) == LOW && direction == LOW) ||
        (digitalRead(limitSwitch2Pin) == LOW && direction == HIGH)) {

      // reverse direction
      direction = !direction;
      digitalWrite(dirPin, direction);

      // unpress switch
      for (int i = 0; i < 50; i++) {  // step count
        singleStep();
      }

      delay(200);  // pause
      break;
    }

    singleStep(); // continue
  }
}

void singleStep() {
  digitalWrite(stepPin, HIGH);
  delayMicroseconds(900);
  digitalWrite(stepPin, LOW);
  delayMicroseconds(900);
}

```

Designed and 3D Printed Squeegee Arms

"Bridge Style"



"Hinge Style"

Learn

- The piping system didn’t arrive on time
- 3D print for one of the designs didn’t fit correctly
- Long prints were time limited out of makerspace

Fixing:

- Re-printing failed items
- Attaching tubing together

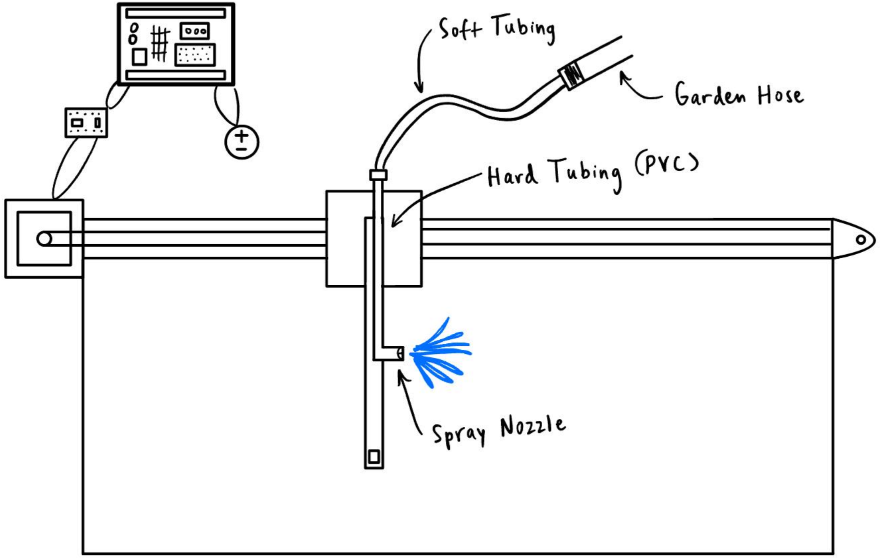
Prototype 5

Define

- Pick “Hinge” or “Bridge” system
- Attach piping system
- Four different pieces, fit and attach together for the final prototype.
- Build housing for arduino, make sure it is spray proof.

Ideate

- Attach all parts together (tubing, squeegee, gantry, window)



Calculate

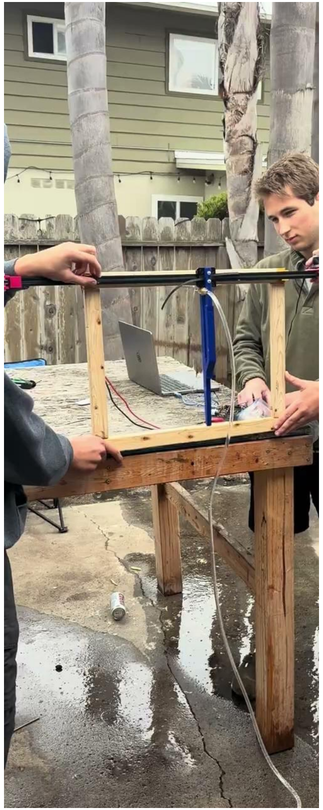
We need to know the speed the water must be sprayed from the gantry in order to cover the whole window. For this we can use conservation of fluid momentum with $Q_{up} = Q_{total}/2$:

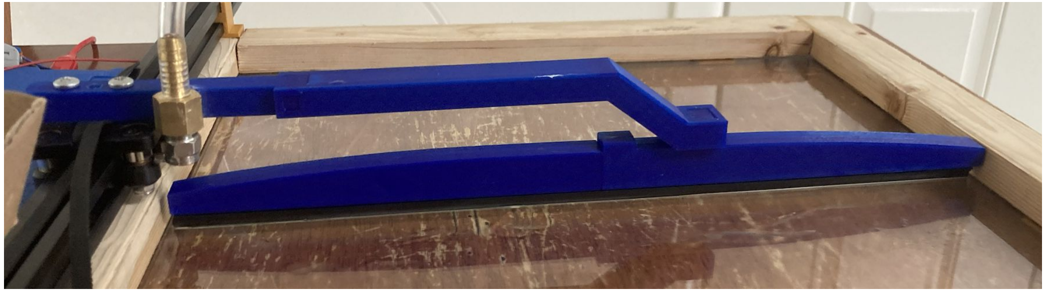
Equating the forces in the y direction, $v_{up} = v_{in}/2$ so:

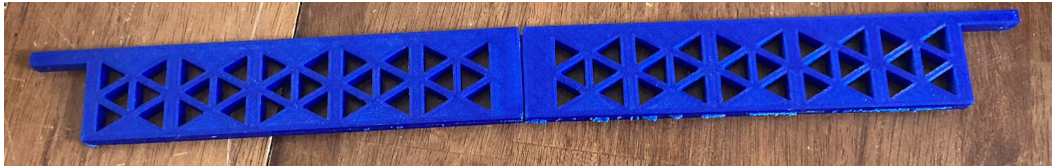
$x = v_{up}^2/(2g)$ so $v_{in} = (8gx)^{½}$

At x = 1 in, v_{in} = 4.6 ft/s

Prototype/Test







Learn

The mechanism worked well:

Overall:

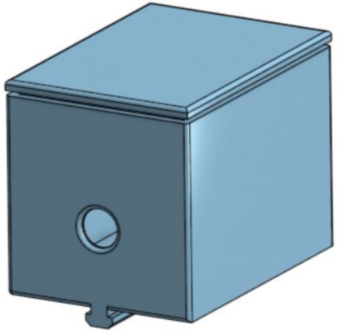
- Cleaned the window back and forth
- Smooth movement between the gantry and V-slot
- Good change in direction
- Enough water to clean the window with minimal friction

Potential Issues:

- Connection between the window and squeegee could be perfected
- Hose could be attached to the frame rather than someone holding it
- Motor mount bends slightly off center, fix could help with better belt movement
- Electronics could be housed in a better fashion (current covered with plastic)

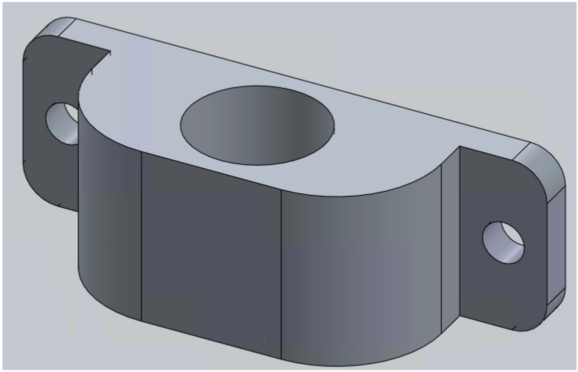
Polish for Final Presentation

Electronic Housing



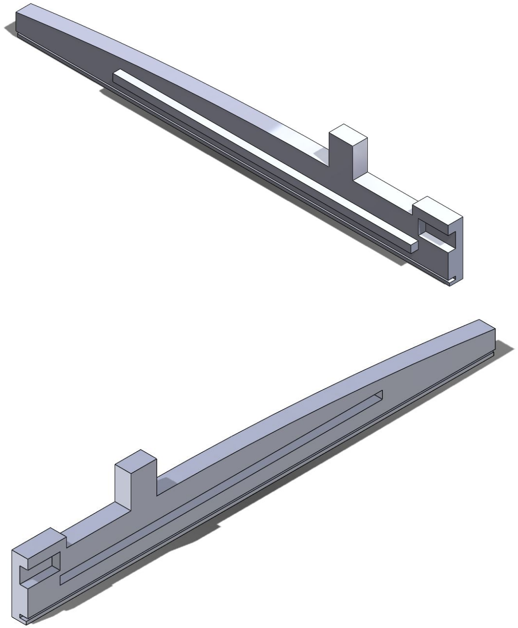
Box Attaching
to T slot

Plumbing Mount



Drilled into frame

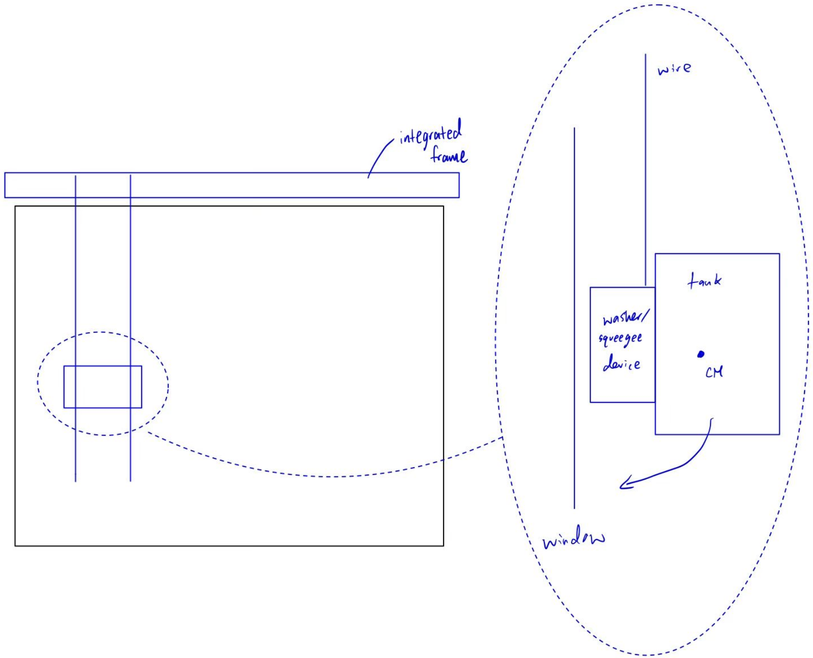
Wiper Reprint



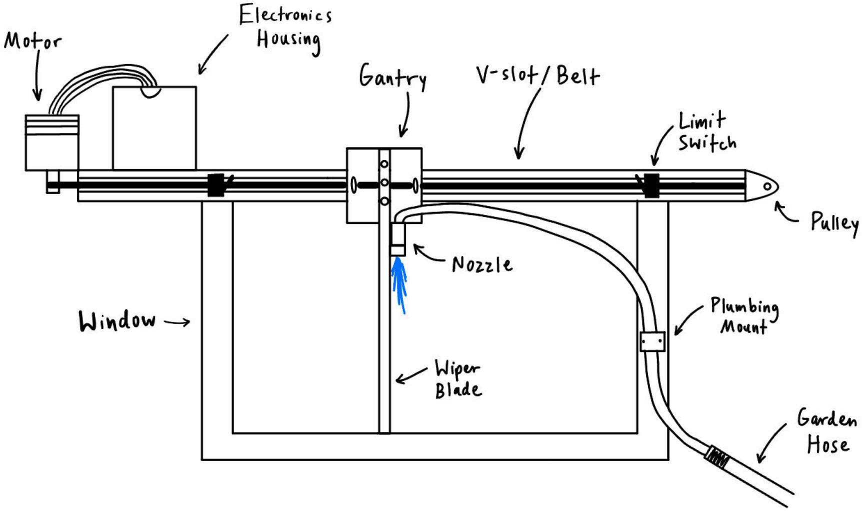
Slight Angle at Tip

Planning

TA Meeting #1 Design



Current Schematic



Biggest differences

Initial Design:

- Intended to move in both the X and Y directions.
- Water system was integrated through a tank mounted on the squeegee.

Final Design:

- Movement limited to the X direction only.
    - Based on Professor Marks' feedback, Y-direction movement was removed due to two flawed options:
        - Ropes and pulleys
            - Led to uneven cleaning.
        - Solid stick mechanism
            - Resulted in an unpolished, messy appearance.
- Water system delivery remained similar in concept (attached to the gantry), but the tank was replaced with a soft tubing system.

What We Learned

- Expect and plan for failures
    - Wrong parts, incorrect measurements, or print issues were common.
- When failures occur, avoid delays by having backup plans or immediately starting rework.
- Identify and source replacements for failed components right away to minimize downtime.
- Plan at least a week ahead, ideally more, to anticipate needs and prevent last-minute issues.
    - Gateway takes a long time
    - Be ready to adapt: items may arrive incorrectly and prototypes may need quick redesigns.

The final design often changed from our original vision at the beginning of each week. We had to learn to adapt and make quick decisions on the spot, a skill we honed over the course of the project.